

```

from google.colab import drive
from google.colab.patches import cv2_imshow # For displaying images in Colab
import os

# 1. Google Drive Mount Karein
drive.mount('/content/drive')

# 2. DeepFace Install Karein (Agar pehle se install nahi hai)
!pip install deepface

# --- CONFIGURATION (Global variables) ---
base_drive_path = "/content/drive/MyDrive"
folder_name = "AI_Attendance" # Corrected typo to match user's existing folder

DB_PATH = os.path.join(base_drive_path, folder_name, "student_db")
LOG_PATH = os.path.join(base_drive_path, folder_name, "attendance_log.csv")

print(f"DB_PATH set to: {DB_PATH}")
print(f"LOG_PATH set to: {LOG_PATH}")

```

Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive", force_remount=True)

Collecting deepface

```

Downloading deepface-0.0.100-py3-none-any.whl.metadata (33 kB)
Requirement already satisfied: requests>=2.27.1 in /usr/local/lib/python3.12/dist-packages (from deepface) (2.32.4)
Requirement already satisfied: numpy>=1.14.0 in /usr/local/lib/python3.12/dist-packages (from deepface) (2.0.2)
Requirement already satisfied: pandas>=0.23.4 in /usr/local/lib/python3.12/dist-packages (from deepface) (2.2.2)
Requirement already satisfied: gdown>=3.10.1 in /usr/local/lib/python3.12/dist-packages (from deepface) (5.2.2)
Requirement already satisfied: tqdm>=4.30.0 in /usr/local/lib/python3.12/dist-packages (from deepface) (4.67.3)
Requirement already satisfied: Pillow>=5.2.0 in /usr/local/lib/python3.12/dist-packages (from deepface) (11.3.0)
Requirement already satisfied: opencv-python>=4.5.5.64 in /usr/local/lib/python3.12/dist-packages (from deepface) (4.13.0.92)
Requirement already satisfied: tensorflow>=1.9.0 in /usr/local/lib/python3.12/dist-packages (from deepface) (2.20.0)
Requirement already satisfied: keras>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from deepface) (3.13.2)
Requirement already satisfied: Flask>=1.1.2 in /usr/local/lib/python3.12/dist-packages (from deepface) (3.1.3)
Collecting flask-cors>=4.0.1 (from deepface)
  Downloading flask_cors-6.0.2-py3-none-any.whl.metadata (5.3 kB)
Collecting mtcnn>=0.1.0 (from deepface)
  Downloading mtcnn-1.0.0-py3-none-any.whl.metadata (5.8 kB)
Collecting retina-face>=0.0.14 (from deepface)
  Downloading retina_face-0.0.17-py3-none-any.whl.metadata (10 kB)
Collecting fire>=0.4.0 (from deepface)
  Downloading fire-0.7.1-py3-none-any.whl.metadata (5.8 kB)
Collecting gunicorn>=20.1.0 (from deepface)
  Downloading gunicorn-26.0.0-py3-none-any.whl.metadata (5.4 kB)
Collecting lightphe>=0.0.15 (from deepface)
  Downloading lightphe-0.0.25-py3-none-any.whl.metadata (14 kB)
Collecting lightds>=0.0.3 (from deepface)
  Downloading lightds-0.0.3-py3-none-any.whl.metadata (7.3 kB)
Requirement already satisfied: python-dotenv>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from deepface) (1.2.2)
Requirement already satisfied: termcolor in /usr/local/lib/python3.12/dist-packages (from fire>=0.4.0->deepface) (3.3.0)
Requirement already satisfied: blinker>=1.9.0 in /usr/local/lib/python3.12/dist-packages (from Flask>=1.1.2->deepface) (1.9.0)
Requirement already satisfied: click>=8.1.3 in /usr/local/lib/python3.12/dist-packages (from Flask>=1.1.2->deepface) (8.4.0)
Requirement already satisfied: itsdangerous>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from Flask>=1.1.2->deepface) (2.2.0)
Requirement already satisfied: Jinja2>=3.1.2 in /usr/local/lib/python3.12/dist-packages (from Flask>=1.1.2->deepface) (3.1.6)
Requirement already satisfied: MarkupSafe>=2.1.1 in /usr/local/lib/python3.12/dist-packages (from Flask>=1.1.2->deepface) (3.0.2)
Requirement already satisfied: Werkzeug>=3.1.0 in /usr/local/lib/python3.12/dist-packages (from Flask>=1.1.2->deepface) (3.1.8)
Requirement already satisfied: BeautifulSoup4 in /usr/local/lib/python3.12/dist-packages (from gdown>=3.10.1->deepface) (4.13.0)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from gdown>=3.10.1->deepface) (3.29.0)
Requirement already satisfied: packaging in /usr/local/lib/python3.12/dist-packages (from gunicorn>=20.1.0->deepface) (26.2)
Requirement already satisfied: absl-py in /usr/local/lib/python3.12/dist-packages (from keras>=2.2.0->deepface) (1.4.0)
Requirement already satisfied: rich in /usr/local/lib/python3.12/dist-packages (from keras>=2.2.0->deepface) (13.9.4)
Requirement already satisfied: namex in /usr/local/lib/python3.12/dist-packages (from keras>=2.2.0->deepface) (0.1.0)
Requirement already satisfied: h5py in /usr/local/lib/python3.12/dist-packages (from keras>=2.2.0->deepface) (3.16.0)
Requirement already satisfied: optree in /usr/local/lib/python3.12/dist-packages (from keras>=2.2.0->deepface) (0.19.1)
Requirement already satisfied: ml-dtypes in /usr/local/lib/python3.12/dist-packages (from keras>=2.2.0->deepface) (0.5.4)
Collecting lightecc (from lightds>=0.0.3->deepface)
  Downloading lightecc-0.0.6-py3-none-any.whl.metadata (14 kB)
Requirement already satisfied: sympy in /usr/local/lib/python3.12/dist-packages (from lightds>=0.0.3->deepface) (1.14.0)
Requirement already satisfied: joblib>=1.4.2 in /usr/local/lib/python3.12/dist-packages (from mtcnn>=0.1.0->deepface) (1.5.3)
Collecting lz4>=4.3.3 (from mtcnn>=0.1.0->deepface)
  Downloading lz4-4.4.5-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl.metadata (3.8 kB)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas>=0.23.4->deepface) (2.9.0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas>=0.23.4->deepface) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas>=0.23.4->deepface) (2026.7)
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests>=2.27.1->deepface) (3.4.0)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests>=2.27.1->deepface) (3.15)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests>=2.27.1->deepface) (2.3.0)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests>=2.27.1->deepface) (2025.11.12)

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```

from IPython.display import display, Javascript
from google.colab.output import eval_js
from base64 import b64decode

# 3. take_photo() function define karein (Updated for continuous capture and stop option)
def take_photo(filename='photo.jpg', quality=0.8):
    js = Javascript('''
        async function takePhoto(quality) {
            const div = document.createElement('div');
            const captureButton = document.createElement('button');
            captureButton.textContent = 'Capture Attendance';
            div.appendChild(captureButton);

            const doneButton = document.createElement('button');
            doneButton.textContent = 'Done (Stop Camera)';
            div.appendChild(doneButton);

            const video = document.createElement('video');
            video.style.display = 'block';
            const stream = await navigator.mediaDevices.getUserMedia({video: true});

            document.body.appendChild(div);
            div.appendChild(video);
            video.srcObject = stream;
            await video.play();

            // Resize the output to fit the video element.
            google.colab.output.setIframeHeight(document.documentElement.scrollHeight, true);

            return new Promise((resolve) => {
                captureButton.onclick = async () => {
                    const canvas = document.createElement('canvas');
                    canvas.width = video.videoWidth;
                    canvas.height = video.videoHeight;
                    canvas.getContext('2d').drawImage(video, 0, 0);
                    // Don't stop stream, just capture and return data
                    const imageData = canvas.toDataURL('image/jpeg', quality);
                    resolve({action: 'capture', data: imageData});
                };

                doneButton.onclick = () => {
                    stream.getVideoTracks()[0].stop();
                    div.remove();
                    resolve({action: 'done', data: null});
                };
            });
        }
    ''')

    display(js)
    result = eval_js('takePhoto({})'.format(quality))

    if result['action'] == 'capture' and result['data']:
        binary = b64decode(result['data'].split(',')[1])
        with open(filename, 'wb') as f:
            f.write(binary)
        return filename, 'captured'
    elif result['action'] == 'done':
        return None, 'done'
    return None, 'error' # Should not happen, but for safety

```

```

import os

# Make sure this matches the DB_PATH from your first cell
DB_PATH = "/content/drive/MyDrive/AI_Attendance"

if os.path.exists(DB_PATH):
    print(f"✅ Success: The database path '{DB_PATH}' exists!")
else:
    print(f"❌ Error: The database path '{DB_PATH}' still does not exist. Please double-check your Google Drive or the path in")

✅ Success: The database path '/content/drive/MyDrive/AI_Attendance' exists!

```

```

import os
import pandas as pd
from datetime import datetime

```

```

import cv2
from google.colab.patches import cv2_imshow
from deepface import DeepFace

# --- STEP 1: PATH VERIFICATION (Already set in f9729cb1) ---
base_drive_path = "/content/drive/MyDrive"
folder_name = "AI_Attendance" # Make sure this matches your folder name

DB_PATH = os.path.join(base_drive_path, folder_name, "student_db")
LOG_PATH = os.path.join(base_drive_path, folder_name, "attendance_log.csv")

def run_ai_attendance():
    # Folder existence check
    if not os.path.exists(DB_PATH):
        print(f"❌ Error: Database path nahi mila!")
        print(f"Mene is path par dhunda: {DB_PATH}")
        print("\n💡 Tip: Drive icon (left side) pe click करके folder pe right-click karein aur 'Copy Path' karke yahan paste kare")
        return

    # Get list of all registered students
    registered_students = [d for d in os.listdir(DB_PATH) if os.path.isdir(os.path.join(DB_PATH, d))]
    if not registered_students:
        print("⚠️ Warning: No student folders found in DB_PATH. Please add student folders with images.")
        return

    print("📷 Camera is ready. Click 'Capture Attendance' for each student, or 'Done (Stop Camera)' to finish.")

    while True:
        try:
            filename, status = take_photo()

            if status == 'done':
                print("✅ Attendance session ended by user.")
                break # Exit the loop if user clicks 'Done'
            elif status == 'error':
                print("⚠️ Error capturing photo. Please try again.")
                continue
            elif status == 'captured' and filename is None:
                print("No photo captured, but camera is still active.")
                continue

            print("🔍 AI is processing... Please wait.")

            recognized_student_name = None
            recognized_student_emotion = "N/A"

            # 2. DeepFace Search
            results = DeepFace.find(img_path=filename,
                                   db_path=DB_PATH,
                                   enforce_detection=False,
                                   model_name="VGG-Face",
                                   distance_metric="cosine")

            if len(results) > 0 and not results[0].empty:
                # If a face is recognized, extract details
                identity_path = results[0]['identity'][0]
                recognized_student_name = os.path.basename(os.path.dirname(identity_path))

                # 3. Emotion Analysis for the recognized face
                analysis = DeepFace.analyze(img_path=filename,
                                           actions=['emotion'],
                                           enforce_detection=False)
                recognized_student_emotion = analysis[0]['dominant_emotion']

                print(f"✅ Recognized: {recognized_student_name}")
                print(f"😊 Mood: {recognized_student_emotion}")
            else:
                print("❌ Face not recognized for this capture. Marking all registered students as Absent.")

            # --- Attendance Logging Logic ---
            now = datetime.now()
            current_date = now.strftime("%Y-%m-%d")
            current_time = now.strftime("%H:%M:%S")

            all_attendance_entries = []

            for student in registered_students:
                status = 'Absent'

```

```

        mood = 'N/A'
        if student == recognized_student_name:
            status = 'Present'
            mood = recognized_student_emotion

        entry = {
            'Date': current_date,
            'Time': current_time,
            'Name': student,
            'Attendance Status': status,
            'Mood': mood
        }
        all_attendance_entries.append(entry)

    # Convert to DataFrame and update CSV
    new_attendance_df = pd.DataFrame(all_attendance_entries)

    if os.path.exists(LOG_PATH):
        existing_log_df = pd.read_csv(LOG_PATH)
        df_final = pd.concat([existing_log_df, new_attendance_df], ignore_index=True)
    else:
        df_final = new_attendance_df

    df_final.to_csv(LOG_PATH, index=False)
    print(f"✅ Attendance log updated successfully at {LOG_PATH}")

    # --- Display Image ---
    img = cv2.imread(filename)
    if recognized_student_name:
        cv2.putText(img, f"{recognized_student_name} (Present, {recognized_student_emotion})", (20, 45),
                    cv2.FONT_HERSHEY_SIMPLEX, 1, (0, 255, 0), 2) # Green for present
    else:
        cv2.putText(img, "Unrecognized Face (All Absent)", (20, 45),
                    cv2.FONT_HERSHEY_SIMPLEX, 1, (0, 0, 255), 2) # Red for unrecognized
    cv2.imshow(img)

    print("Ready for next student or click 'Done (Stop Camera)' to finish.")

except Exception as e:
    print(f"⚠️ Error during attendance process: {e}")
    print("Attempting to continue or waiting for user input...")

print(f"Final attendance log saved at: {LOG_PATH}")

# Run the AI Attendance System
run_ai_attendance()

```

📷 Camera is ready. Click 'Capture Attendance' for each student, or 'Done (Stop Camera)' to finish.

🔍 AI is processing... Please wait.

26-05-30 03:14:44 - Found 0 newly added image(s), 0 removed image(s), 4 replaced image(s).

Finding representations: 0% | 0/4 [00:00<?, ?it/s] Downloading...

From: https://github.com/serengil/deepface_models/releases/download/v1.0/vgg_face_weights.h5

To: /root/.deepface/weights/vgg_face_weights.h5

26-05-30 03:14:48 - 📁 vgg_face_weights.h5 will be downloaded from https://github.com/serengil/deepface_models/releases/download/v1.0/vgg_face_weights.h5

```

0%|          | 0.00/580M [00:00<?, ?B/s]
2%||         | 12.1M/580M [00:00<00:04, 116MB/s]
6%|||        | 36.2M/580M [00:00<00:02, 189MB/s]
12%|||       | 67.1M/580M [00:00<00:02, 241MB/s]
16%|||      | 91.8M/580M [00:00<00:02, 213MB/s]
20%|||     | 117M/580M [00:00<00:02, 225MB/s]
24%|||    | 140M/580M [00:00<00:02, 190MB/s]
29%|||   | 168M/580M [00:00<00:01, 214MB/s]
33%|||  | 191M/580M [00:00<00:01, 217MB/s]
37%||| | 214M/580M [00:01<00:01, 194MB/s]
42%|||  | 244M/580M [00:01<00:01, 221MB/s]
46%|||  | 267M/580M [00:01<00:01, 225MB/s]
51%|||  | 294M/580M [00:01<00:01, 235MB/s]
55%|||  | 318M/580M [00:01<00:01, 212MB/s]
60%|||  | 346M/580M [00:01<00:01, 229MB/s]
64%|||  | 370M/580M [00:01<00:01, 185MB/s]
67%|||  | 390M/580M [00:01<00:01, 177MB/s]
72%|||  | 416M/580M [00:02<00:00, 197MB/s]
75%|||  | 437M/580M [00:02<00:00, 178MB/s]
79%|||  | 457M/580M [00:02<00:00, 161MB/s]
83%|||  | 480M/580M [00:02<00:00, 176MB/s]
87%|||  | 504M/580M [00:02<00:00, 190MB/s]
90%|||  | 525M/580M [00:02<00:00, 192MB/s]
94%|||  | 546M/580M [00:02<00:00, 197MB/s]
100%|||  | 580M/580M [00:06<00:00, 90.8MB/s]

```

Finding representations: 100% | 4/4 [00:24<00:00, 6.19s/it]

26-05-30 03:15:09 - There are now 4 representations in ds_model_vggface_detector_opencv_aligned_normalization_base_expand_0.pk

26-05-30 03:15:09 - Searching photo.jpg in 4 length datastore

26-05-30 03:15:10 - find function duration 27.89318060874939 seconds

Downloading...

From: https://github.com/serengil/deepface_models/releases/download/v1.0/facial_expression_model_weights.h5

To: /root/.deepface/weights/facial_expression_model_weights.h5

26-05-30 03:15:11 - 📁 facial_expression_model_weights.h5 will be downloaded from https://github.com/serengil/deepface_models/releases/download/v1.0/facial_expression_model_weights.h5

100% | 5.98M/5.98M [00:00<00:00, 175MB/s]

🟢 Recognized: Rimsha Abbas Khan

👤 Mood: neutral

🟢 Attendance log updated successfully at /content/drive/MyDrive/AI_Attendance/attendance_log.csv



Ready for next student or click 'Done (Stop Camera)' to finish.

🟢 Attendance session ended by user.

Final attendance log saved at: /content/drive/MyDrive/AI_Attendance/attendance_log.csv

◆ Gemini

```
import pandas as pd
import os
```

```
# Ensure LOG_PATH is defined (it should be from previous cells)
```

```
# LOG_PATH = os.path.join("/content/drive/MyDrive", "AI_Attendance", "attendance_log.csv")

if os.path.exists(LOG_PATH):
    attendance_df = pd.read_csv(LOG_PATH)
    print("Current Attendance Log:")
    display(attendance_df)
else:
    print(f"The attendance log file does not exist yet: {LOG_PATH}")
    print("Please run the `run_ai_attendance()` function first to generate the log.")
```

Current Attendance Log:

	Date	Time	Name	Mood	Attendance	Status	
0	2026-05-28	04:39:19	Rimsha Abbas Khan	sad		NaN	
1	2026-05-28	04:58:06	Rimsha Abbas Khan	sad		Present	
2	2026-05-28	04:58:06	Muhammad Abbas Khan	NaN		Absent	
3	2026-05-28	04:58:06	Haris Abbas	NaN		Absent	
4	2026-05-28	04:58:06	Faiza Afzal	NaN		Absent	
5	2026-05-28	05:05:04	Rimsha Abbas Khan	sad		Present	
6	2026-05-28	05:05:04	Muhammad Abbas Khan	NaN		Absent	
7	2026-05-28	05:05:04	Haris Abbas	NaN		Absent	
8	2026-05-28	05:05:04	Faiza Afzal	NaN		Absent	
9	2026-05-29	03:28:12	Rimsha Abbas Khan	neutral		Present	
10	2026-05-29	03:28:12	Muhammad Abbas Khan	NaN		Absent	
11	2026-05-29	03:28:12	Haris Abbas	NaN		Absent	
12	2026-05-29	03:28:12	Faiza Afzal	NaN		Absent	
13	2026-05-29	03:30:42	Rimsha Abbas Khan	NaN		Absent	
14	2026-05-29	03:30:42	Muhammad Abbas Khan	NaN		Absent	
15	2026-05-29	03:30:42	Haris Abbas	NaN		Absent	
16	2026-05-29	03:30:42	Faiza Afzal	NaN		Absent	